RESEARCH EVIDENCE / LGTO

Frustrated magnetism. Sub-kelvin cooling.

Govind Unnikrishnan | IISER Thiruvananthapuram
$Li_3Gd_3Te_2O_{12}$ (LGTO) | Supervisor: Dr. Deepshikha Jaiswal-Nagar

SCIENTIFIC QUESTION

Can field-induced criticality and entropy accumulation enhance adiabatic demagnetization cooling in a frustrated rare-earth garnet?

MY CONTRIBUTION

Prepared polycrystalline samples; submitted powder for XRD and assessed phase purity using FullProf/Rietveld refinement.

Performed PPMS VSM, heat-capacity and ADR measurements, and made sample contacts.

Participated in zero-field and field-dependent heat-capacity measurements down to 100 mK using a ^{3}He-^{4}He dilution refrigerator at UGC-DAE CSR, Indore.

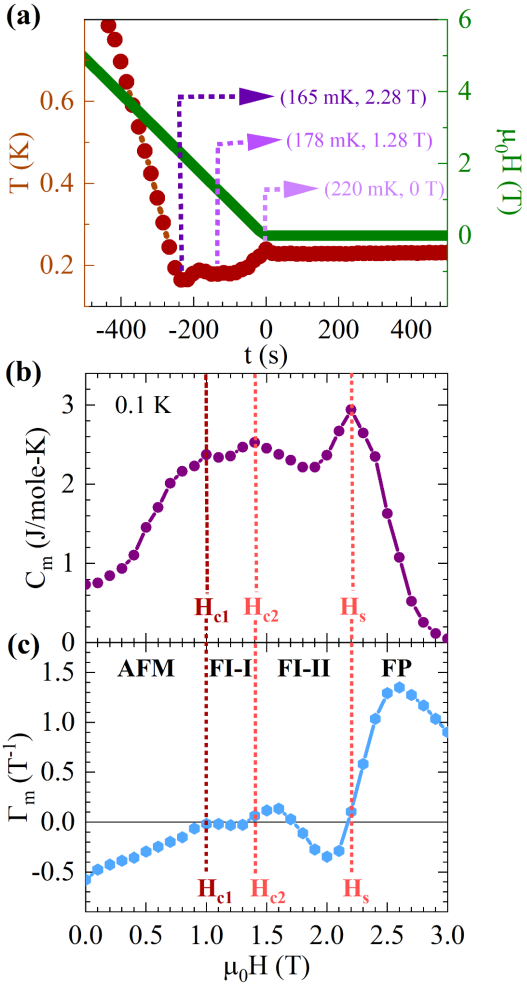
Analysed the data and assisted in writing the manuscript.

MEASURED OUTCOMES

165 mK minimum during demagnetization, at **2.28 T**.
>12 hours below 1 K in the reported setup, starting from 1.8 K and 7 T.
Magnetic ordering near **250 mK**; field-dependent heat capacity measured at **100 mK**.

WHAT I LEARNED

Connecting thermodynamic measurements with direct cooling traces reveals how the magnetic entropy landscape affects refrigeration. Heat leaks, thermal contact and the measurement protocol also matter when interpreting performance.



Reading the evidence

(a) Direct ADR cooling trace.
(b) Field-dependent magnetic heat capacity at 0.1 K, measured using dilution refrigeration.
(c) Magnetic Grüneisen analysis of the cooling response, with the phase labels proposed in the study.

The combined results support field-induced critical behaviour and entropy accumulation near critical fields. They motivate the use of frustrated magnets for cooling below their ordering temperature.

Source: D. Ranaut et al., "A Frustrated Garnet $Gd_3Te_2Li_3O_{12}$ for High-Performance sub-Kelvin Adiabatic Demagnetization Refrigeration." Accepted for publication. Figure 6 reproduced from the authors' manuscript.

govind21@iisertvm.ac.in | govindjocose006.github.io/#lgto